TR-Core: Autonomous Symbolic Induction for Structural-OOD Generalization

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The Problem: Why Neural Networks Fail at Structural OOD

| Limitation | Consequence |

|---|---|

Memorization over composition | Fails on valid but unseen concept combinations

Black-box opacity | Cannot audit, debug, or repair decisions

Data hunger | Requires massive labeled datasets

Static knowledge | Cannot adapt without full retraining

No causal mechanism | Correlation mistaken for composition

Structural-OOD Example

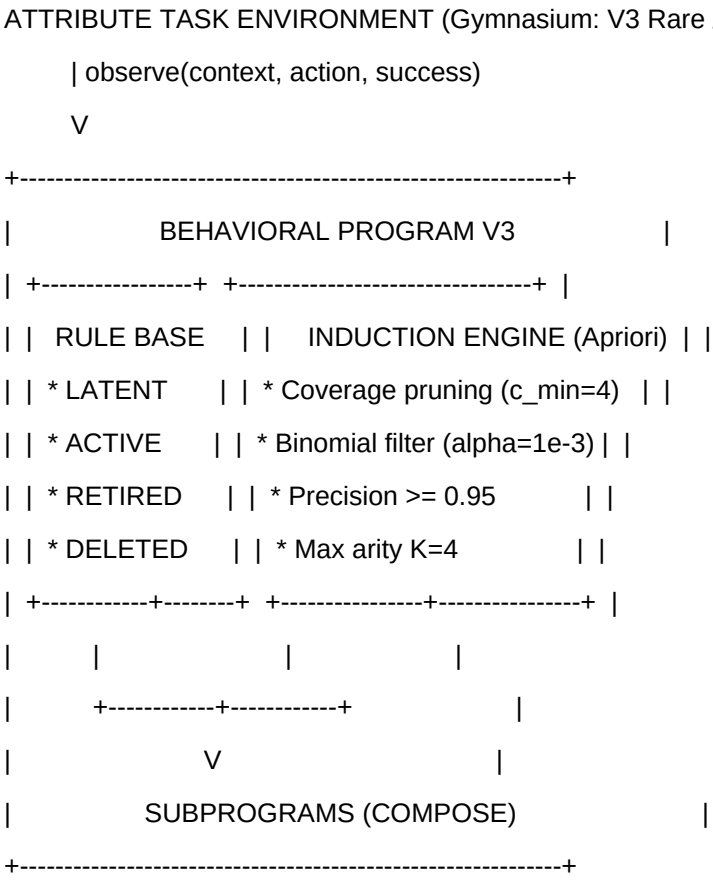
Learned: calm AND guard IMPLIES approach

Held-out (never seen): calm AND guard AND night IMPLIES approach

Neural nets: ~0.25 accuracy (chance)

Required: Compositional generalization via COMPOSE

TR-Core Architecture



7 Development Operations: CREATE, GENERALIZE, SPECIALIZE, SPLIT, COMPOSE, MODIFY, DELETE

Phase 3: Rare Compositional Rules (8 attrs, 4,608 contexts)

Model	Balanced Accuracy (Main/Repl)
TR-Core (full_bp)	0.784 / 0.803
ValueDListML	0.813 / 0.766
ValueDecisionTree	0.752 / 0.763
ValuekNN	0.702 / 0.710
ValueRandomForest	0.659 / 0.671
Logistic Regression	0.200 / 0.200
ab_no_compose / null_bp	0.200 / 0.200

Rule Recovery: 68% exact/subset | Behavioral FDR = 0.36

Phase 4: Structural-OOD (Held-out triple, zero training support)

Model	Held-out Accuracy (Main/Repl)
TR-Core (full_bp)	0.958 / 1.000
ValueDecisionTree	0.977 / 0.985
ValueDListML	0.910 / 0.995
ValueRandomForest	0.844 / 0.935
ValuekNN	0.253 / 0.311
ab_no_compose / null_bp	0.000 / 0.000

Pair rule recovery: calm AND guard IMPLIES approach recovered 24/24 seeds

Ablation Studies: Causal Evidence

Ablation	Phase 3 Balanced Acc	Phase 4 Held-out Acc
full_bp (COMPOSE enabled)	0.784 / 0.803	0.958 / 1.000
ab_no_compose (COMPOSE disabled)	0.200 / 0.200	0.000 / 0.000
null_bp (randomized outcomes)	0.200 / 0.202	0.000 / 0.000
chance baseline	0.199 / 0.205	0.251 / 0.236

Conclusion: COMPOSE is the causal mechanism for structural-OOD generalization.

Without it, the system collapses to random guessing. With it, it achieves near-ceiling OOD accuracy.

Formal Guarantees & Infrastructure

Formal Analysis (Proven)

| Property | Bound |

|---|---|

Worst-case time | O(N*L + L^K)

Expected time (pruning) | O((1+beta*L)^K)

Space | O(N + L + R*K)

Type I error (per candidate) | <= alpha = 0.001 (exact binomial)

Professional Infrastructure

- Gymnasium interface: TRCore-AttributeTaskV3-v0, V4-v0
- Configuration: Hydra (sweep-ready YAML)
- Pipeline: DVC (14 stages: data to exp to analysis to figures)
- Statistics: Permutation tests (20k), Cohen's dz, bootstrap CI
- Reproducibility: 24 seeds x 2 replications, fixed seeds, DVC tracking
- Package: pip install -e . via pyproject.toml

Conclusions & Next Steps: Alan

What TR-Core Proves

1. Compositional generalization without neural networks -- symbolic induction + exact statistics suffices
2. COMPOSE is necessary and sufficient for structural-OOD -- ablations isolate it as causal mechanism
3. Exact statistical control enables tractable search -- binomial filter + coverage floor tames combinatorial explosion
4. White-box by construction -- every decision traceable to a rule with provenance

Next Steps: Alan (Embodied AGI Prototype)

Phase	Goal	Timeline
1. Perception	Pixels to discrete attributes (Slot Attention + DINOv2)	Months 1-3
2. Hierarchical Composition	Rules to Skills to HTN Programs (library learning)	Months 4-7
3. Lifelong Memory	Episodic to Semantic to Procedural consolidation	Months 8-11
4. Embodiment	Sim2Real on Unitree Go1 / ALOHA (CBF-MPC safety)	Months 12-18

TR-Core is the verified symbolic core. Alan builds the organism around it.

Contact & Links

Repository: github.com/Mardon-programm/tabula-rasa-core

Alan (Next Phase): github.com/Mardon-programm/alan

Paper (LaTeX): docs/tr_core_paper.tex

Formal Analysis: docs/apriori_formal_analysis.tex

Demo: streamlit run web/app.py -> Phase 4 Demo

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